

Manufacturing Process

Week-8

1. Introduction to sheet metal-Show You tube videos on sheet metal operations and Applications
2. Explain sheet metal materials -Standard Gauges of sheet -Specification of Sheet
3. Explain different Sheet metal operations

Principle of sheet metal working

Sheet metal working is generally associated with press machines and press working. Press working is a chipless manufacturing process by which various components are produced from sheet metal. The thickness of metal varies from 0.1 mm to 10 mm. Press machine consists of a frame which supports a ram and bed and a mechanism for operating the ram. The ram is equipped with punch whereas die block is attached to the bed. The punch and die block assembly is called die or die-set. Die block is a stationary part which contains die cavity and punch is a moving part which enters in the die cavity. During the operation, metal sheet is kept on the die block and the punch moves downward. The punch forces the metal sheet into the die cavity, hence metal sheet will form the shape of the die cavity. There is always some clearance between the punch and die block. On the press machine, various operation can be performed and all the operations are done at room temperature.

Advantages, disadvantages and Application of Sheet Metal Working:

Advantages

- Sheet metal working is associated with a press machine, on which number of operations can be performed.
- Metal sheets of less thickness can be formed into various shapes.
- The components produced by sheet metal working are of low cost.
- Production rate of press machine is very high.
- The process does not require skilled labour.

Disadvantages

- Sheet metal working is only used for mass production.
- The cost of die is very high.
- Initial cost of the press machine is also high.
- Metals of thickness more than 10 mm are difficult to form.
- The operation produces more noise and vibrations.

Applications

The components produced by sheet metal working are as follows:

- Press parts are widely used in the automobile (bikes, cars, trucks, buses, etc.) industry.
- Vehicle parts like doors, roofs, fuel tanks, front guards, etc. can be produced.
- Aircraft industry, radio and telephone industry, electrical parts, etc.

specifications of metal sheets

The specifications of metal sheets are given in terms of their gauge numbers, length and width. Gauge number represents the thickness of metal sheets. The higher the gauge number, the smaller the thickness.

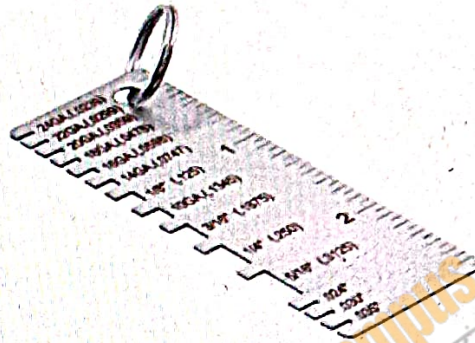


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Sheet metal thickness is commonly described using a gauge system, where a higher gauge number indicates a thinner sheet, and the most common range used is 26 gauge (thinner) to 7 gauge (thicker).

Gauge System:

A sheet metal gauge/gage is an indication of the standard thickness of specific sheet metal materials, such as steel, aluminum, and copper. The larger the gauge number, the thinner the metal (not applicable for zinc).



A metal gauge chart will contain the gauge number and the corresponding thickness of certain metal, in inches or millimeters.

Common Range:

The most commonly used sheet metal gauges range from 26 gauge (thinner) to 7 gauge (thicker).

Gauge and Thickness:

The gauge number doesn't directly correspond to a specific thickness in inches or millimeters, but a gauge chart can be used to determine the actual thickness.

Example:

18 gauge steel measures approximately 0.0478 inch or 1.214 millimeters.

Copper Gauge Chart

Gauge	Thickness	
	Inch	mm
7	0.180	4.572
8	0.165	4.191
9	0.148	3.759
10	0.134	3.404
11	0.120	3.048
12	0.109	2.769
14	0.083	2.108
16	0.065	1.651
18	0.049	1.245
20	0.035	0.889
22	0.028	0.711
24	0.022	0.559
26	0.018	0.457
28	0.014	0.356
30	0.012	0.305



Visual Watermark

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Metals used in sheet metal working

There are different types of metals used in sheet metal work in the form of sheets and plates. Some of the important sheet metals are as follows:

1. Black Iron

Black iron or uncoated sheet carries no artificial coating on its surface, but is cheaper than other types of metal sheets. Components made from this type of metal are pans, tanks, cabinets, almirahs, stove, pipes, etc.

2. Galvanised Iron

Its soft iron sheet carries zinc coating on its surface which makes the surface good looking and rust resistant. Components made from this type of metal are storage tanks, buckets, heating ducts, furnaces, gutters, pans, trunks, etc.

3. Aluminium sheets

Due to low strength of aluminium sheets they are not used in their pure form, hence a suitable amount of silicon manganese, copper and iron are added. It offers high resistance to corrosion and abrasion. They are used in the manufacture of aeroplane bodies, kitchenware and cabinets, doors, windows and building work, electrical appliances, etc.

4. Copper sheets

Copper sheets are costlier but offers good resistance to corrosion and are relatively good in appearance. They are reddish in colour, highly ductile and malleable. They are used in applications like radiators of automobiles, heating appliance, gutters, hoods and components in chemical plants.

5. Stainless steel

Stainless steel offers high resistance to corrosion and exhibits a bright surface. It is used in the manufacture of food containing equipment, dairy equipment, food processing plant, chemical plant, etc.

6. Tin plates

Tin plates are used for those iron sheets which are coated with pure tin. Tin plates are used for making good containers, containers for cooking oils and ghee, cans, etc.



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Metal Cutting Operations

In sheet metal cutting operations, the metal gets sheared hence these operations are also called shearing operations. In these operations, the metal sheet is stressed beyond its ultimate strength. Metal cutting operations include following operations:

1. Blanking
2. Punching (Piercing)
3. Notching
4. Perforating
5. Slitting
6. Lancing
7. Shaving

1. Blanking

Blanking is the cutting operation of a flat metal sheet and the article punched out is known as blank. Blank is the required product of the operation and the metal left behind is considered as a waste.

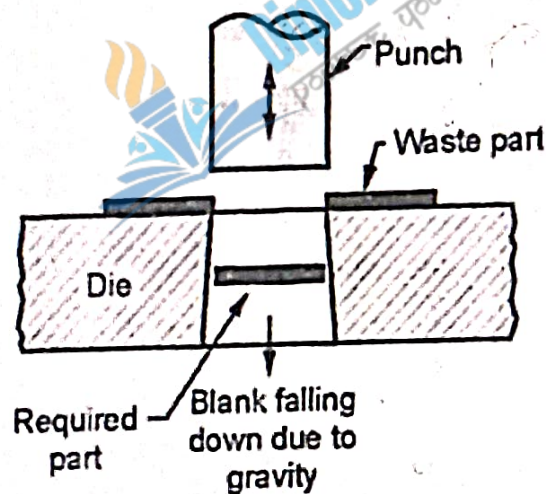


Fig. 4.3 : Blanking



Manufacturing Process

2. Punching (Piercing)

It is the cutting operation with the help of which holes of various shapes are produced in the sheet metal. It is similar to blanking; only the main difference is that, the hole is the required product and the material punched out to form a hole is considered as a waste.

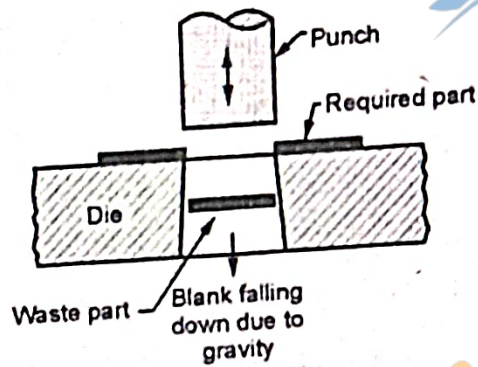


Fig. 4.4 : Punching (Piercing)

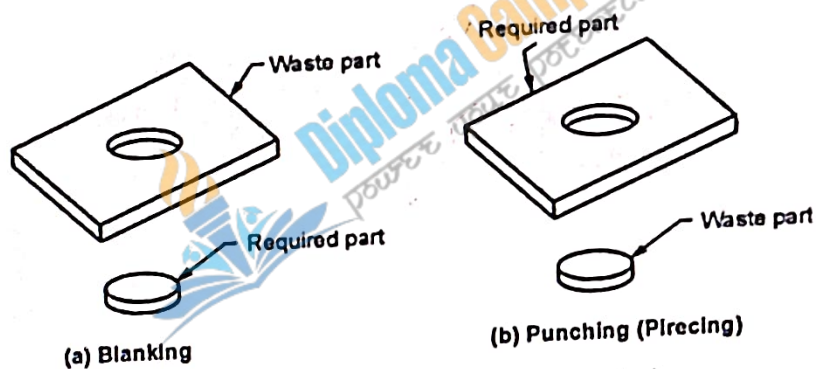


Fig. 4.5 : Blanking and punching (piercing)

3. Notching

It is similar to blanking operation, but in this full surface of punch does not cut the metal. In this operation, metal pieces are cut from the edges of a sheet.

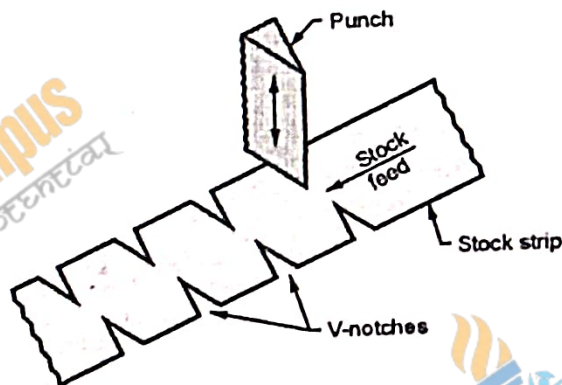


Fig. 4.6 : Notching



Visual Watermark

4. Perforating

It is similar to piercing but the difference is that, to produce holes the punch is not of round shape. In this process, multiple holes which are very small and close together are cut in the sheet metal.

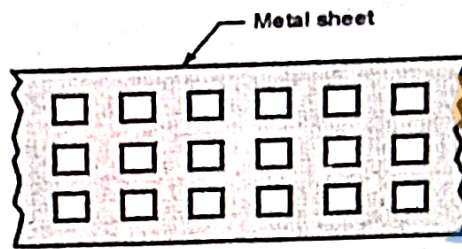


Fig. 4.7 : Perforating

5. Slitting

It is the operation of making an unfinished cut through a limited length only.

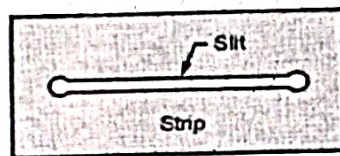


Fig. 4.8 : Slitting

6. Lancing

In this operation, there is a cutting of sheet metal through a small length and bending this small cut portion downwards.



Fig. 4.9 : Lancing

7. Shaving

This operation is used for cutting unwanted excess material from the periphery of a previously formed workpiece. In this process very small amount of material is removed.

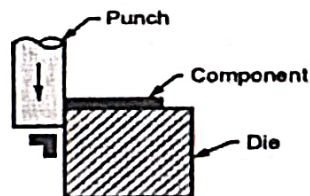


Fig. 4.10 : Shaving

Metal forming operations

In metal forming operations, the sheet metal is stressed below the ultimate strength of the metal. In these operations, no material is removed hence there is no wastage. Metal forming operations include following operations:

1. Bending
2. Drawing
3. Embossing
4. Forming
5. Coining (Squeezing)

1. Bending

It is a metal forming operation in which the straight metal sheet is transformed into a curved form. In bending operations, the sheet metal is subjected to both tensile and compressive stresses. During the operation, plastic deformation of material takes place beyond its elastic limit but below its ultimate strength.

2. Drawing

In this operation, punch forces a sheet metal blank to flow plastically into the clearance between the punch and die. Finally, the blank takes a shape of cup.

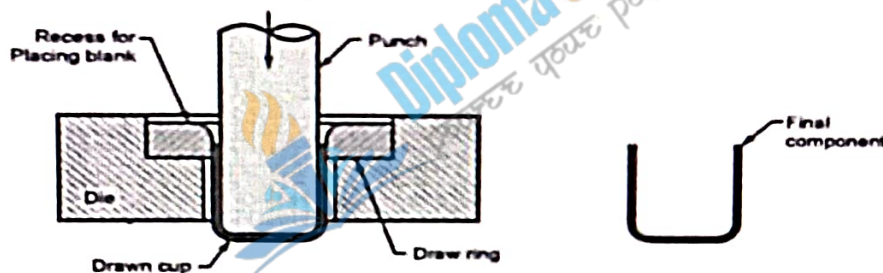


Fig. 4.19 : Drawing

3. Embossing

With the help of this operation, specific shapes or figures are produced on the sheet metal. It is used for decorative purposes or giving details like names, trademarks, specification, etc. on the sheet metal.

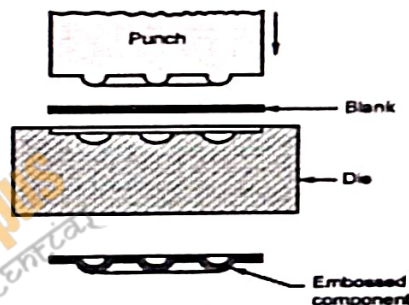


Fig. 4.20 : Embossing

4. Forming

In forming operation, sheet metal is stressed beyond its yield point so that it takes a permanent set and retains the new shape. In this process, the shape of punch and die surface is



directly reproduced without any metal flow. This operation is used in the manufacturing of door panels, steel furniture, air-craft bodies, etc.

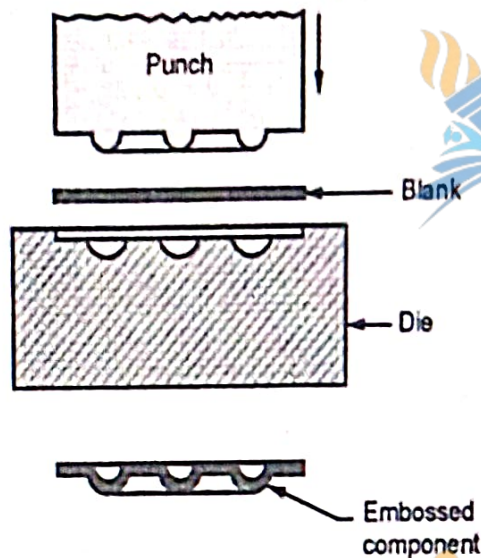


Fig. 4.20 : Embossing

5. Coining (Squeezing)

In coining operation, the metal having good plasticity and of proper size is placed within the punch and die and a tremendous pressure is applied on the blank from both ends.

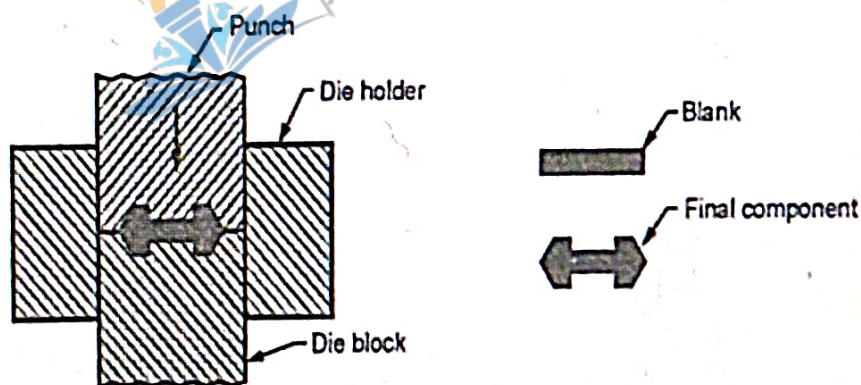


Fig. 4.22 : Coining

Under severe compressive loads, the metals flow in the cold state and fill up the cavity of the punch and die. This operation is used in the manufacturing of coins, medals, ornamental parts, etc.